



Society of Petroleum Engineers

SPE-228956-MS

ADNOC Onshore Fastest ERD Well Case Study, Utilizing New Solutions, Resulting Time Reduction & Efficiency

Shouq Al Memari, L. Gerardo, A. Sumaida, R. Alrahama, A. Aboudraz, G. Salem, A. Hamouda, and A. Elshahat, ADNOC Onshore; W. Fares, M. Abouzaid, and M. Elsayfy, Halliburton

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/228956-MS](https://doi.org/10.2118/228956-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper outlines the development of a gas field in a mature carbonate reservoir via an onshore Extended Reach Drilling (ERD) well in Abu Dhabi, featuring a 24,000-foot lateral section. The operator aimed to achieve a drilling rate of 1 mile per day. However, both the landing and lateral sections presented significant drilling and geological challenges, including subsurface uncertainty, rig limitations, high downhole temperatures, and Effective Circulating Density (ECD) management. In addition, maintaining 100% reservoir contact required frequent well path adjustments via manual geosteering, introducing performance variability and human error. These challenges constrained the rate of penetration (ROP) and increased overall well costs. This paper presents a customized drilling and subsurface solution combined with a downhole drilling automation system that enabled 100% reservoir contact and saved 4.6 drilling days.

A new Intelligent Rotary Steerable System (RSS) was deployed in combination with an Ultra High Torque Motor to maintain consistent weight on bit (WOB) and differential pressure (>500 PSI while drilling) along with compatible bit to achieve target ROP. Deep Reservoir Mapping tools were critical in navigating the thin carbonate reservoir, enabling real-time identification of formation dip changes and reservoir boundaries above and below the wellbore, contributing to higher ROP. Autonomous downhole directional drilling tools allowed continuous well-path adjustments, enhancing placement precision within the reservoir. Integration with automated geosteering software reduced decision-making cycle time from ~15 minutes to <5 minutes. The closed-loop automation also minimized RSS setting changes, promoting smoother wellbores and facilitating efficient deployment of the production Bottom Hole Assembly (BHA), while reducing invisible lost time. Vibration mitigation was achieved through the combined use of Motorized RSS and Anti-Stick Slip tools.

The well reached total depth (TD) at 34,127 feet Measured Depth (MD) successfully with 23,095 feet of 8½" lateral section drilled, and 100% reservoir contact achieved. A record onshore daily footage of 4,076 feet was attained in the UAE. This marked the global debut of the new 6¾" Motorized RSS. Integration of the autonomous downhole directional drilling system with the Motorized RSS and Deep Reservoir Mapping enabled precise control throughout the lateral section. The solution reduced RSS control interventions,

improved ROP by 47%, and significantly enhanced the speed and accuracy of geosteering versus offset wells.

The successful integration of automated and customized drilling technologies represents a transformative step in ERD well execution, significantly reducing well time and increasing drilling efficiency. The scalable and repeatable nature of this autonomous downhole drilling solution offers a robust framework for future field development in similarly complex environments.

Introduction

Efficient reservoir access through long horizontal wells in tight carbonate environments requires a delicate balance between drilling performance, real-time trajectory control, and geological placement accuracy. Well A located in onshore field of Abu Dhabi, was designed as a gas producer with a planned horizontal drain of ~24,000 feet through a narrow 12-ft TVD window within the AA reservoir. The complexities of this project included rig capacity limitations, high-temperature conditions, and the high uncertainty with formation dipping and fluid contacts (notably the gas-oil contact, or GOC). The objective was to execute this well with enhanced drilling efficiency, reduced tortuosity, and 100% productive reservoir exposure. Furthermore, maintaining a high rate of penetration (ROP) while executing manual or semi-automated geosteering decisions. The wired motorized Rotary Steerable System is considered to be suitable solution to address the rig capacity limitations, however, still has its own limitations and reliability issues to maintain the required Weight On Bit (WOB) and differential pressure. In addition, traditional methods of manual Geo-Steering and reactive decision-making proved insufficient in similar environments, prompting the adoption of an integrated solution combining high-torque RSS, deep resistivity mapping and autonomous trajectory control.

This manuscript focuses on three main components of the execution strategy:

- Pre-Well Modelling and BHA Planning
- Real-Time Execution with Integrated Technologies
- Post-Well Analysis and Model Validation

New Intelligent high-performance rotary steerable system Solution

The new high-performance rotary steerable system is a motorized intelligent rotary steerable system, powered by a high-performance mud motor to deliver superior drilling performance and maximize the value of wellbore assets. The addition of a high-performance mud motor above the RSS extends the system's operational range. The new RSS features a high transmission electro-magnetic (EM) signal hop that enables efficient communication between the RSS and the rest of the BHA and supports prolonged operation in complex environments. [Figure 1](#). In addition to high mechanical specifications, modular design, high-strength materials and connections, high RPM capability up to 400 rpm and it's compatible with the automated downhole directional drilling solution.

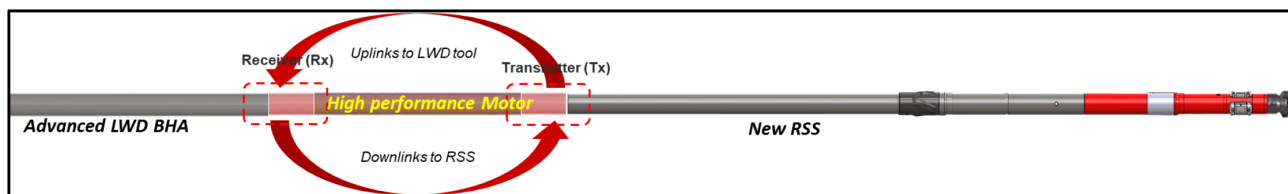


Figure 1—New Intelligent high-performance rotary steerable system layout.

LWD Reservoir Mapping solution

In the last ten years, induction-based resistivity technology has undergone a significant transformation, particularly in its role in modern well placement strategies. A key milestone in this evolution was the emergence of Ultra-Deep Azimuthal Resistivity (UDAR) tools combined with 1D inversion algorithms, which redefined the scope and impact of resistivity logging. This technological leap enabled a shift from conventional geosteering—focused solely on maintaining borehole trajectory within reservoir boundaries—to a broader approach that includes proactive reservoir mapping and real-time structural decision-making such as geostopping before exiting the pay zone. Today's UDAR systems offer depth-of-investigation (DOI) capabilities that extend from Shallow inversions of 25 feet to beyond 200 feet of Deep inversions, allowing operators to visualize geological structures well ahead and laterally away from the wellbore. This wide-ranging DOI has proven particularly valuable in mapping structural dips, fluid contacts and formation boundaries in large, heterogeneous carbonate systems. However, the challenge remains when targeting thinly bedded reservoirs—those with vertical thicknesses less than 25 feet—where ultra-deep mapping lacks the resolution needed to guide accurate placement. In such settings, deeper-sensing tools often blur formation details, making precise geosteering difficult and increasing the risk of wellbore exit or reduced reservoir exposure. To address these thin-bed challenges, a more refined solution has emerged through the pairing of 1D inversion processing with Deep Azimuthal Resistivity (DAR) tools. These tools are designed with a different architecture compared to their ultra-deep counterparts. A defining characteristic of DAR tools is their co-located transmitter and receiver arrays within the same collar, a design that enhances measurement sensitivity and improves tool reliability in dynamic drilling environments. This compact configuration is especially beneficial in extended-reach and directional wells, where collar spacing and borehole stability are critical constraints. (Fares et al. 2024). What sets the DAR system apart is its advanced transmitter design, which employs state-of-the-art electrical components to generate clean, high-frequency electromagnetic signals. These signals are captured using orthogonally arranged transmitter-receiver pairs, supported by precision-engineered coils and low-noise electronics. The tool architecture allows for dual-array measurements—Deep and Base arrays—providing frequency options ranging from 62.5 kHz to 250 kHz, adaptable to various formation properties. These raw components serve as the foundation for real-time inversion modeling, enabling detailed stratigraphic interpretation and responsive geosteering adjustments.

Figure 2

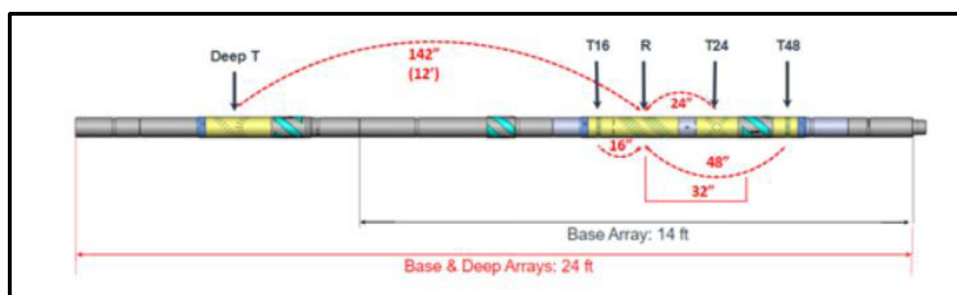


Figure 2—Tool Layout of Deep-Azimuthal-Resistivity (DAR) tool (Shehab et al., 2024).

The combination of advanced hardware and high-resolution 1D inversion makes DAR tools particularly effective in detecting layer boundaries, resistivity contrasts and formation dips within closely spaced geological intervals. By delivering enhanced vertical resolution at moderate depths of investigation, this technology bridges the gap between conventional shallow azimuthal tools and UDAR systems, offering a balanced solution for complex reservoir environments.

In essence, the evolution of azimuthal resistivity tools—culminating in the integration of 1D inversion with DAR systems—has significantly broadened the toolkit available for precise well placement. It

empowers drilling teams to maintain optimal reservoir contact, reduce formation exit risks and make smarter, data-driven decisions in both simple and geologically complex formations.

Drilling Extended Drilling Reach (ERD) wells plays a crucial role in the development of mature oil fields. These intricate operations necessitate accurate control of the wellbore trajectory to meet objectives during various stages such as landing, lateral drilling, and geosteering. There is always a potential risk of collision with already existing wells or laterals, which mandates comprehensive real-time anti-collision and survey management systems. In normal practice, directional drillers (DD) have relied heavily on their deep understanding and experience of BHA behavior in specific geological settings. DD would adjust drilling parameters and modify assembly configurations as required. However, this method often resulted in deviations from the intended wellbore trajectory and led to performance inconsistencies due to possibility in human errors aspect.

This paper advocates for a groundbreaking approach that involves the integration of automated drilling and geosteering software to manage the downhole BHA, thus significantly reducing the necessity for human intervention. Autonomous drilling demands a sophisticated software solution capable of delivering real-time, highly accurate calculations to control the downhole RSS. This enables continuous corrections to the wellpath, ensuring it remains within the designated target zone. The proposed system is seamlessly integrated with automated geosteering software, which remarkably reduces the decision-making time.

Furthermore, the integration between the automated drilling and geosteering software minimizes the need for frequent adjustments to the RSS settings. Such a system ensures a smoother wellbore trajectory and reduces Invisible Lost Time (ILT). The implementation of this automated system represents a significant advancement in drilling technology, offering a new level of operational efficiency and consistency that was previously unattainable.

Automated downhole directional drilling solution

The Automated downhole directional drilling solution is designed to operate the drilling BHA while reducing the need for constant manual intervention and allowing operators to focus on higher-level tasks. Advanced sensors continuously monitor the drilling environment, providing crucial data that informs the system's decision-making processes. Anti-collision scanning technology ensures that the drill avoids existing wells and other obstacles, minimizing the risk of accidents and enhancing overall safety.

The autonomous drilling platform requires an intelligent RSS which is designed with the latest technology to enable accurate well placement and complex downhole calculations. The RSS can accurately steer the well up to 400 rpm while measuring the tool face 1,000 times per second. It is integrated with automated drilling advisory commands based on physics-based models, a BHA digital twin, and machine learning from realtime data analytics:

- Steering advisory commands help plan, project the well path, optimize wellbore trajectory, and avoid hazards and collisions
- Vibration advisory commands advise on optimal RPM ranges to manage downhole vibration and allow for better directional control by automating the drilling process,

The geosteering solution helped in mapping subsurface formations using imaging techniques and real-time data from downhole sensors. It identifies conductive zones and steers the drill bit away from potential hazards. The downhole automation platform solution then sends the projected well path for the autonomous directional drilling system to adjust tool settings and correct the well path to meet target specifications. The process ensures wellbore placement to maximize reservoir contact and enhance hydrocarbon recovery.

The autonomous directional drilling platform combined with geosteering software is designed to ease drilling in the complex reservoirs. It uses real-time data analytics and advanced sensors to perform anti-collision scanning, optimize well trajectory, and reduce drilling time. The system provides precise control,

enabling operators to make informed, data-driven decisions and adjust drilling parameters dynamically to improve performance. The use of advanced imaging techniques allows for a more detailed understanding of subsurface geology. By utilizing data from various sources such as seismic surveys, electromagnetic measurements, and resistivity logs, the system creates a comprehensive model of the formation. This model helps in making informed decisions about the direction and depth of the drilling. The goals include dynamic downhole uncertainty adaptation for real-time well steering, instantaneous real-time decision-making with adaptive calibration algorithms, and consistent and repeatable well delivery.

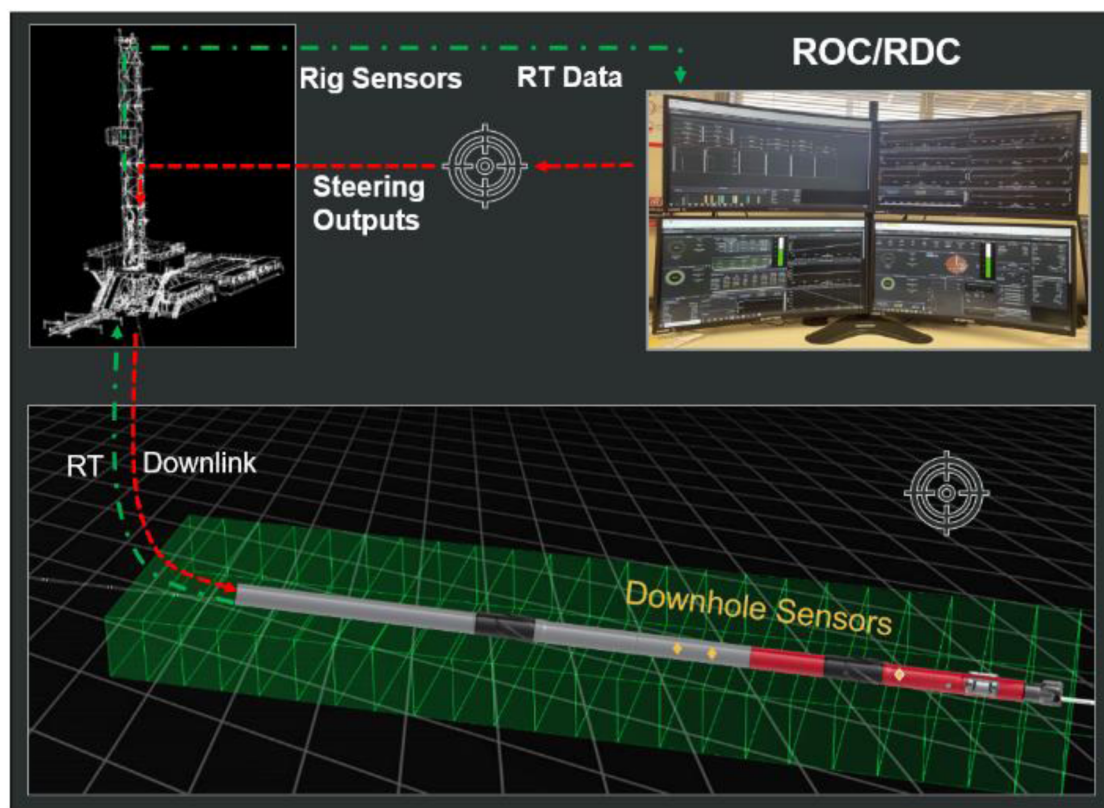


Figure 3—Autonomous Drilling Platform.

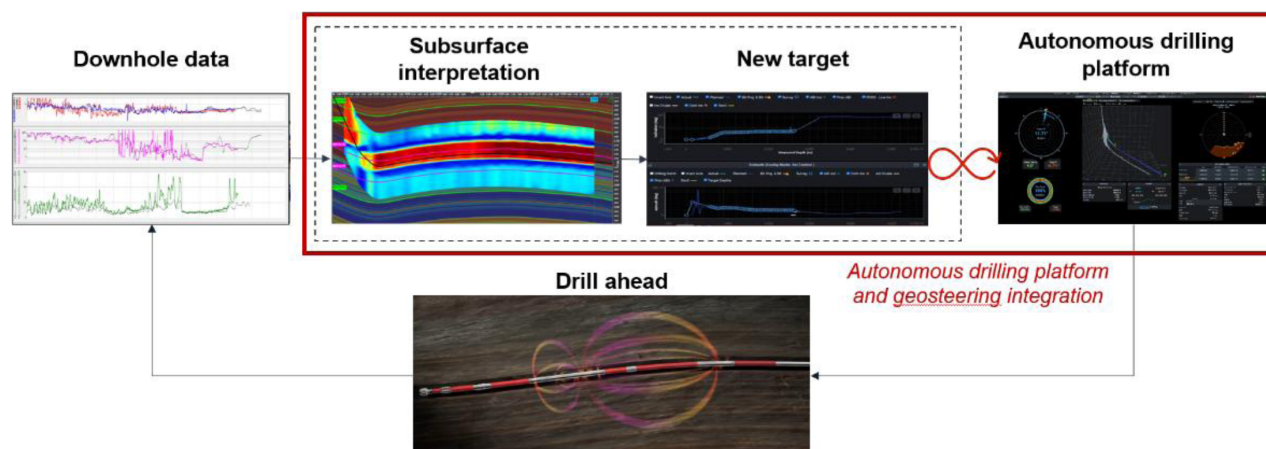


Figure 4—Workflow for Automated Well Placement.

Pre-Well Modelling

A successful well execution hinges on meticulous planning and pre-well modeling that integrates geology, petrophysics, and real-time drilling technologies. For Well A, this process began with the compilation and interpretation of offset data from well Well 1-01D (Fig.2). This offset was selected due to their comparable structural positioning and log characteristics. Petrophysical evaluation highlighted AA RESERVOIR as the target interval with consistent reservoir signatures: porosity between 13–16 pu, resistivity in the 8–15 ohm-m range, and low gamma ray counts (6–8 API). These characteristics served as the foundation for geosteering decisions and real-time log comparison. Geological modeling was carried out using state of the art inversion modelling software. These platforms enabled simulation of synthetic resistivity, density, and gamma ray logs under different structural scenarios. Importantly, they supported the integration of 1D and 2D inversions of deep azimuthal resistivity data to identify directional changes in bedding and detect structural anomalies in advance of the bit. The modeling phase revealed the importance of a resistive sublayer, termed AB RESERVOIR-DNS, located approximately 4–5 ft below the productive reservoir. This sublayer provided a critical geosteering reference, enabling consistent DTB (distance-to-boundary) tracking. Modeling also addressed expected fault interactions (Fig.5). A seismic cross-section overlaid with log-derived structure maps indicated at least two significant fault planes intersecting the planned trajectory, requiring inclination adjustments to maintain wellbore stability and reservoir contact. To avoid undesirable contacts, particularly with the underlying A1 Reservoir or the gas-oil contact at 8870 ft TVDss (designated as NO GO ZONE), a set of boundary conditions were defined. These included inclination thresholds, escalation protocols for formation dip deviations and predefined workflows for Remote Operation Control Centre (ROC)-based geosteering updates. Overall, the pre-well model served as the operational blueprint for maintaining geosteering precision across the lateral section.

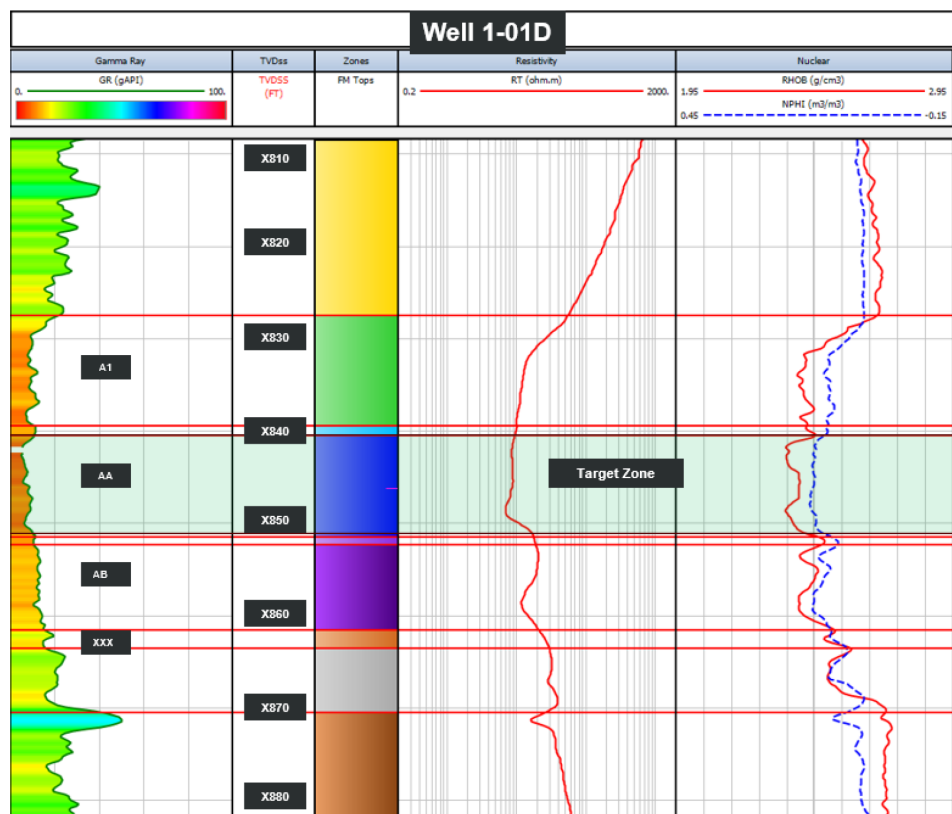


Figure 5—Offset Well 1-01D with triple combo logs identifying the target zone.

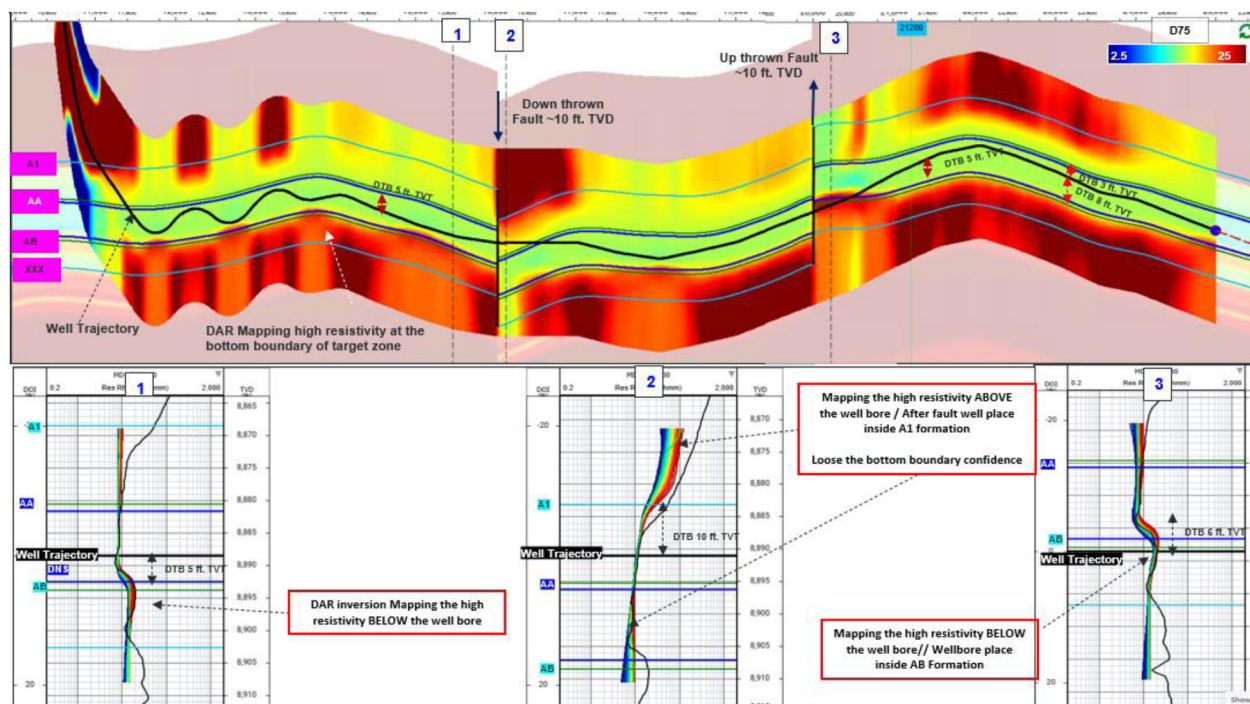


Figure 6—Pre-well 1D inversion modeling using deep azimuthal resistivity for Well A, considering fault scenarios within the original structural framework.

BHA Design and Planning

The BHA was carefully engineered to optimize drilling efficiency, wellbore quality, and trajectory control throughout the 8½" extended-reach lateral section. Considering the length of the 8½" lateral section length along with well TD at 34127 feet MD along with reservoir pressure uncertainties and to avoid Stuck with chemical Radioactive source of LWD tools, it has been decided to drill the 8½" lateral section in 2 bit runs as follow:

First bit run: Using New Intelligent high-performance rotary steerable system along with high performance mud motor, near Bit Gamma ray, Deep Azimuthal resistivity for reservoir mapping/propagation resistivity, Density and Neutron porosity. Figure 8. It was planned to drill 10,000 feet lateral section then stop drilling and pull out of hole to change BHA and run in hole to resume drilling with the radioactive sourceless BHA.

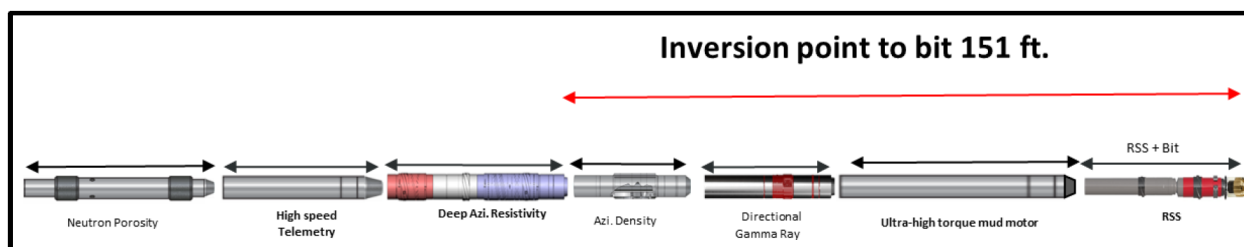


Figure 7—First Bit run BHA: New Intelligent high-performance rotary steerable system along with high performance mud motor, Deep azimuthal resistivity, Density/Neutron porosity schematic bottom hole assembly configuration diagram for 8.5in hole section (RA Source BHA)

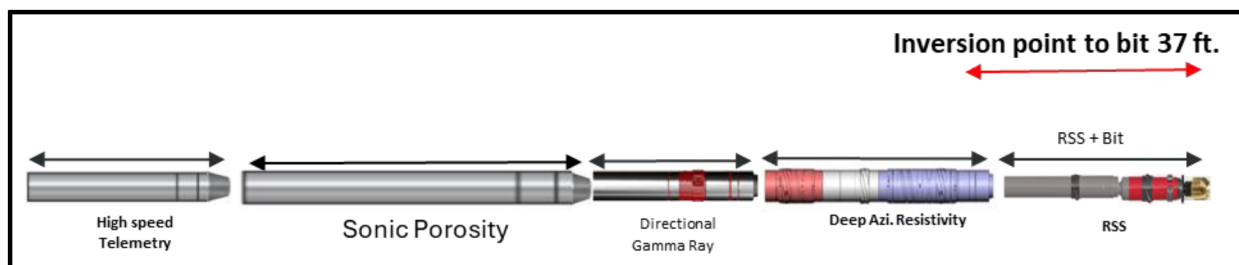


Figure 8—Second Bit Run BHA: Intelligent rotary steerable system along with Deep azimuthal resistivity and Sonic tool schematic bottom hole assembly configuration diagram for 8.5in hole section (RA Sourceless BHA)

Second bit run: Using intelligent rotary steerable system along with near Bit Gamma ray, Deep Azimuthal resistivity for reservoir mapping/propagation resistivity and acoustic sonic (sourceless porosity). Figure 8. It was decided to continue drilling with this BHA till planned TD at 34,127 feet MD.

A detailed evaluation was conducted between two directional drive systems: a newly introduced motorized RSS configuration which integrates a high-torque, seven-stage mud motor with the RSS and the conventional standalone RSS. The New Intelligent high-performance RSS was selected as an enabling technology based on its ability to deliver elevated bit RPM independently of surface rotation. This decoupling allowed for increased cutting efficiency while maintaining lower surface RPM, thereby reducing top drive torque demand, limiting mechanical wear on the drill string, and mitigating torsional vibration. Figure 9. These attributes proved especially beneficial over long horizontal intervals, where surface-driven systems often encounter diminishing energy transfer and reduced drilling responsiveness due to cumulative drag. Although both systems offered acceptable trajectory control capabilities, the New Intelligent high-performance RSS configuration provided enhanced energy availability at the bit, which translated into more consistent ROP and smoother directional corrections. This was particularly advantageous in the reservoir's low-reactivity carbonate lithology, where maintaining bit engagement and drilling momentum is often a challenge. Furthermore, by enabling higher bit RPM without relying solely on surface input, the motorized system supported improved cutting structure efficiency and reduced stick-slip behavior under varied downhole conditions. To address the hole cleaning which is one of the main ERD challenges, it was planned to optimize the mud fluid with 2 new systems to mitigate friction, high torque & drag challenges, assure efficient hole cleaning & excellent ECD management.

Bit selection was driven by offset well analysis and formation-specific performance trends. A six-blade polycrystalline diamond compact (PDC) bit with 16 mm cutters and an extended-gauge pad was selected for its proven ability to balance aggressive rock failure with directional stability. In prior wells, this design minimized spiraling tendencies and mitigated the formation of localized doglegs, both of which are common contributors to wellbore tortuosity and completion-related challenges. The extended gauge also aided in maintaining wellbore smoothness during high build-rate intervals and provided better support for trajectory hold in zones with varying formation dip.

To support accurate geosteering and real-time formation evaluation, the BHA incorporated a layered measurement-while-drilling (MWD) and logging-while-drilling (LWD) sensor suite. Near-bit gamma and density sensors were positioned close to the bit to deliver immediate lithological feedback, enabling timely trajectory adjustments. Additionally, a deep azimuthal resistivity sensor was placed approximately 150 ft behind the bit to provide structural insight well ahead and around the wellbore. This setup allowed the geosteering team to interpret boundary movements and formation dips proactively, particularly in faulted or stratigraphically transitional zones. The combination of the new motorized RSS drive, offset-informed bit design, and strategically sequenced measurement tools resulted in a fit-for-purpose BHA configuration.

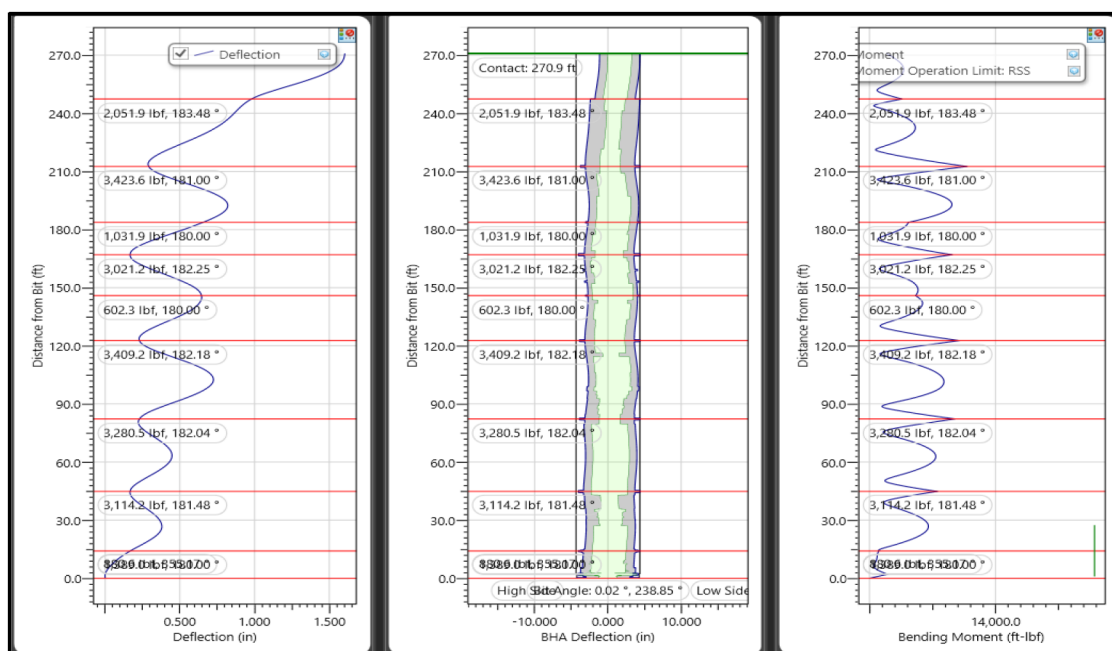


Figure 9—Drilling engineering BHA analysis and simulation.

Real-Time Execution

Real-time drilling commenced from the 9-5/8" casing shoe at 11,050 ft MD with the deployment of a dual-BHA strategy. The first BHA included the new Intelligent high-performance rotary steerable system paired with a high-torque motor, gamma-ray and density/Neutron Porosity tools and Deep Azimuthal Resistivity Tool (DAR). This assembly allowed for precise inclination control and enhanced trajectory smoothing during landing. This integrated system delivered a measurable improvement in drilling performance, including higher sustained ROP, reduced directional dysfunction, and smoother wellbore profiles. It also enabled enhanced reservoir navigation and in-zone placement over the entire interval in the first bit run and exceeded the planned footage and reached 12,123 feet lateral section, setting a new performance benchmark for the field's ERD development strategy. After drilling approximately 12,123 feet MD, the BHA was replaced with a Source-less BHA as planned, incorporating the same DAR with the addition of the Sonic tool replacing the radioactive Neutron Porosity, for operations in environmentally sensitive zones. The DAR was activated in real-time inversion mode with a 20 ft Depth of Investigation (DOI). This setting provided continuous boundary tracking above and below the wellbore, enabling precise placement decisions even under ambiguous log signatures. The Geosteering team, operating from Real-Time Operations Center (ROC), monitored boundary movements, resistivity shifts, and inclination trends to guide steering decisions. From the onset, formation tops appeared deeper than modeled (Figure 10), prompting the team to adjust inclination to 89.8° for optimal reservoir landing. The reservoir entry was confirmed by the emergence of the AB Reservoir-DNS resistive signature in inversion maps. Over the next 23,095 ft, the well was steered dynamically to maintain a position within 4–5 ft from the reservoir's upper boundary. DAR facilitated early fault detection, most notably at 25,843 ft MD, where a sharp resistivity change signaled a structural offset. The Geosteering Engineer used dip interpretations and real-time reservoir mapping inversion to initiate corrective inclination adjustments, returning the trajectory to AA RESERVOIR in minimal footage. This proactive approach minimized reservoir exit events and reduced tortuosity. The automated downhole directional drilling workflow was fully utilized to execute the Geosteering decisions and adjustment to the well trajectory and helped in reducing the number of required Geosteering decisions. Also, it helped to reduce the downlinks by providing a drilling parameters recommendations and optimize the downlinks to achieve required well trajectory adjustment. These reductions in goesteering decisions and downlinks helped

to avoid reaching dense (low porosity) zones and accordingly increase the average ROP compared to offset wells. Mitigating the hole cleaning and to maintain smooth borehole, During the first run, hole-cleaning performance was evaluated through real-time surveillance of torque, drag, and equivalent circulating density (ECD) prior to making each connection—whether short (10 ft), single, or full stand. This methodology provided a reliable means of validating wellbore conditions before progressing with drilling operations. In the second run, utilizing a standalone rotary steerable system (RSS), surface rotary speed was intentionally increased to the 140–150 rpm range. This operational adjustment was designed to enhance cuttings transport efficiency and sustain hole-cleaning effectiveness, thereby mitigating the impact of the extended lateral section. As a significant part of the operational highlights, the below were observed:

- Sustained wellbore placement in the productive zone
- Record-breaking onshore drilling footage of 4,076 ft/day
- 47% ROP improvement vs. offset wells
- Vibration mitigation via optimized RSS settings

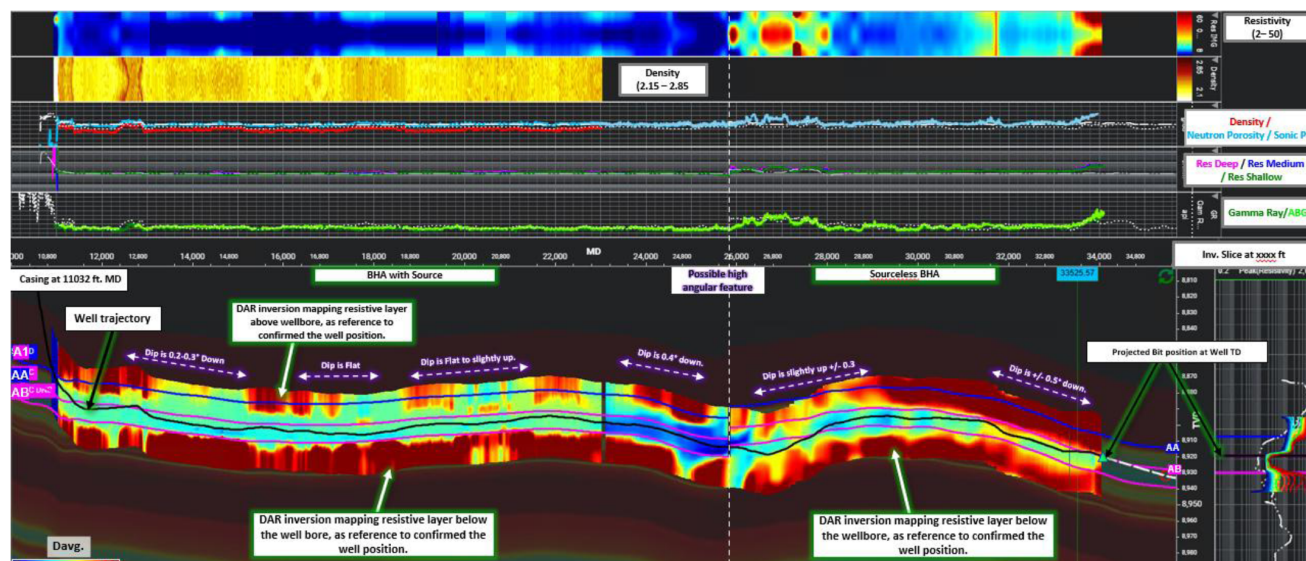


Figure 10—Geosteering cross section for the entire wellbore presenting both bit runs along with Deep azimuthal resistivity 1D reservoir mapping. The well trajectory maintained within the target zone for the entire section with 100% reservoir contact

Results

The well has successfully reached the planned TD at 34,127 feet MD and achieved 23,095 feet of 8 ½" lateral section in 2 planned runs. Table 1. The new Intelligent high-performance rotary steerable system along with high performance mud motor was an efficient solution to push the limit of the ROP, reaching 4,076 feet per day as a new record in onshore wells. The DAR helped to overcome the sensor measurement point (150 feet) away from the bit and provided full reservoir insight and mapped precisely both reservoir boundaries, facilitated a proactive geosteering decisions to maintain the well in target zone and achieve 100% reservoir contact. The automated downhole directional drilling workflow was fully executed in 8 ½" lateral section and achieved all 23 Geosteering decisions without any manual geosteering nor downlinks. Table-1. The automated solution has helped and proved the value to optimize the well trajectory and Geosteering decisions. The Real-time cycle times for Geosteering decisions dropped from ~15 minutes to less than 5 minutes, significantly enhancing operational efficiency and wellbore placement confidence.

Table 1—Performance Summary

Target Zone	Carbonate Reservoir
Total Depth	34,127 feet
Footage	23,095 feet
Number of downlinks	212
On-bottom ROP	157.85 ft/hr
Number of automated geosteering decisions	23

Considering that the footage of this well is longer by 3,000 ft, the number of downlink and Geosteering decisions were normalized per 1000 feet. According to the trial results, the number of downlinks was reduced by 27%, and the number of Geosteering decisions was reduced by 47%. Additionally, the trial results indicated reductions in both aspects compared to the average from offset wells. [Table 2](#)

Table 2—On Bottom ROP Performance Comparison

Subject	Target Well	Offset Well
Footage Drilled	23,095	20008
Downlinks	212	250
	9.17 per 1000ft	12.45 per 1000ft
# GST Decisions	23	33
	0.99 per 1000	1.65 per 1000ft
OB ROP	157.85	84 ft/hr

The on-bottom rate of penetration (ROP) in this well was calculated for a drilling interval of 23,095 feet. The results indicated an increase of 47% in the on-bottom ROP compared to offset well.

The pilot of the automated downhole directional drilling workflow successfully met the four key performance indicators (KPIs) outlined in the performance targets, as summarized in [Table-3](#) below:

Table 3—Key Performance Indicators

Subject	KPI	Target Well
Downlinks	-10-12%	-27%
# GST Decisions	-17%	-40%
OB ROP	-15%	47%
Assurance of RT inc & azi trend Assurance of RT inc & azi trend	80%	100%

Conclusions

The Well A considered as a benchmark ERD well in UAE and globally for integrated digital drilling and geosteering in tight carbonate reservoirs. The combination of the new Intelligent high-performance RSS along with high performance mud motor, autonomous directional drilling, and DAR enabled:

- Achieved planned well TD at 34,127 feet
- 100% reservoir placement in a challenging 23,095 feet of 8 ½" lateral section
- Enhanced ROP by 47% vs. offsets

- Elimination of trajectory-induced tortuosity risks
- Minimization of geological uncertainty through deep resistivity imaging

The combination of automated geosteering and directional drilling systems marks a major step in drilling operations. These innovations enhance drilling accuracy and efficiency, promoting safer and more sustainable resource extraction. By integrating advanced sensors and control algorithms, these systems can precisely navigate the drill bit to the optimal location, reducing the likelihood of hitting non-productive zones and minimizing environmental impact.

The success of Well A demonstrates the value of forward modeling, real-time inversion tracking and automation in delivering high-performance ERD wells. These methodologies are scalable and repeatable across similar carbonate reservoirs in the Middle East and globally.

References

1. Fares, W., Shehab S., Al Marzouqi, S., Al Menhali S., et al., "Reducing Structural Uncertainty and Maximizing Reservoir Contact Through Precise Mapping of 8 Minor Faults Using Advanced Deep Reservoir Mapping Solutions, A Case Study from Onshore Abu Dhabi." Presented at ADIPEC, November 2024, Paper Number: SPE-222944-MS; DOI: <https://doi.org/10.30632/SPWLA-2022-0049>.